

Summations and Expansions

$\sum_{k=1}^n k = \frac{1}{2} n(n+1)$	$\sum_{k=1}^n k^4 = \frac{1}{30} n(n+1)(2n+1)(3n^2+3n-1)$
$\sum_{k=1}^n k^2 = \frac{1}{6} n(n+1)(2n+1)$	$\sum_{k=1}^n k^5 = \frac{1}{12} n^2 (n+1)^2 (2n^2+2n-1)$
$\sum_{k=1}^n k^3 = \frac{1}{4} n^2 (n+1)^2$	$\sum_{k=1}^n k^6 = \frac{1}{42} n(n+1)(2n+1)(3n^4+6n^3-3n+1)$
$\sum_{k=0}^n C^k = \frac{C^{n+1}-1}{C-1} \quad \forall C \in (1, \infty]$	$(x+y)^n = \sum_{i=0}^n \binom{n}{i} x^{n-i} y^i$

Arithmetic Sequence

n_{th} term	$a_n = a + (n-1) \times d$
Sum of the first n terms	$S_n = \frac{n}{2} (2a + (n-1) \times d)$

Geometric Sequence

n_{th} term	$a_n = ar^{n-1}$
Sum of the first n_{th} terms	$S_n = \frac{a(1-r^n)}{(1-r)}$
Sum of the terms infinitely	$S_n = \frac{a}{(1-r)} \quad \forall r < 1$

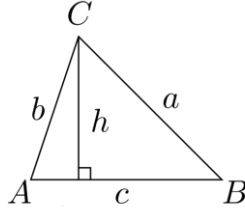
Harmonic Sequence

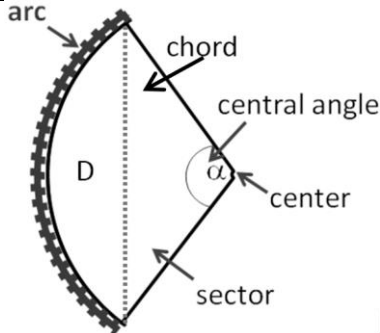
n_{th} term	$a_n = \frac{1}{a + (n-1) \times d}$
Sum of the first n_{th} terms	$S_n = \frac{1}{d} \ln\left(\frac{2a + (2n-1) \times d}{2a-d}\right)$

Trig. Identities

$\sin(A \pm B) = \sin(A)\cos(B) \pm \cos(A)\sin(B)$	$\sin(2A) = 2\sin(A)\cos(A)$
$\cos(A \pm B) = \cos(A)\cos(B) \mp \sin(A)\sin(B)$	$\cos(2A) = \cos^2(A) - \sin^2(A)$
$\tan(a \pm b) = \frac{\tan(a) \pm \tan(b)}{1 \pm \tan(a)\tan(b)}$	$\tan(2A) = \frac{2\tan(A)}{(1-\tan^2(A))}$

Misc. Formulas

Law of sines	$\frac{a}{\sin(A)} = \frac{b}{\sin(B)} = \frac{c}{\sin(C)} = 2r$	
Law of cosines	$c^2 = a^2 + b^2 - 2ab\cos(C)$	
Major diagonal in a reg. polygon		$L \times \sin\left(\frac{\left[\frac{n}{2}\right] \times \pi}{n}\right)$
Area from medians		$\frac{4}{3} \sqrt{sm(sm-ma)(sm-mb)(sm-mc)}$
dot(a, b)		$\ a\ \ b\ \cos(\theta)$

cross(a, b)		$\ a\ \ b\ \sin(\theta)$
Lattice points between two points		$\gcd(\Delta x , \Delta y) + 1$
Polygon area		$\frac{1}{2} \left \sum_{i=1}^n \text{cross}(p_i, p_{(i+1)\%n}) \right $
Polygon area (Pick's Theorem)		Interior Pts. + $\left(\frac{\text{Boundary Pts.}}{2}\right) - 1$
Regular polygon area		$\frac{1}{2} nr^2 \sin\left(\frac{2\pi}{n}\right)$; r = radius circumcircle
		$\frac{nL^2}{4 \tan\left(\frac{\pi}{n}\right)}$
Regular polygon angle		$\frac{(n - 2) \times 180}{n}$
Circular sector area	$\frac{1}{2} r^2 \times \alpha$	
	$\pi r^2 \times \frac{\alpha}{360}$	
D shape area	$\frac{1}{2} r^2 (\alpha - \sin(\alpha))$	
Chord length	$2 \pi r \times \sin\left(\frac{\alpha}{2}\right)$	
Arc length	$2\pi r \times \frac{\alpha}{360}$	
Spanning trees complete bipartite graph K_{nm}		$m^{n-1} \times n^{m-1}$
Cyclic quad. area		$\sqrt{(s - a)(s - b)(s - c)(s - d)}$; $s = \frac{a + b + c + d}{2}$
Radius of circle inscribed in a triangle		$\frac{2 \times \text{Area}}{\text{perimeter}}$
Spanning trees in complete graph		n^{n-2}
Euler's formula for planar graphs		Vertices – Edges + Faces = 2
Derangements*		$\text{der}(n) = (n - 1) \times (\text{der}(n - 1) + \text{der}(n - 2))$
Distinct binary trees with n nodes		$\text{Cat}(n) = \frac{C(2n, n)}{n + 1}$
Balanced expressions of n pair of parentheses		
Ways n + 1 factors can be parenthesized*		
Triangulation of n + 2 convex polygon		
Monotonic paths along edges in n x n grid		
		
Triangulation of a Convex Polygon		Monotonic Paths

*Derangements: permutations of length n with no value positioned correctly

*n = 3, factors = {a,b,c,d} , we have (ab)(cd),a(b(cd)),((ab)c)d,(a(bc))(d),a((bc)d)